MODE-S Wind — Windshear detection v1

The six browser algorithms of the Windshear page (Pairwise, Gradient, Energy, Rate, Baseline, Kinematic), in use May – 24 October 2026 · legacy documentation · Otso Laakso (OH2GAX) · github.com/oh2gax/mode_s_wind

Status: retired on 24 October 2026. From version v.24102026 the Windshear page no longer runs these algorithms. They were replaced by one server-side detector (v2): an along-track headwind *ramp* detector on the aircraft's own BDS 5,0 replies (TAS − GS), scaled by ramp length ($\Delta V \cdot (1\ \text{km/L})^{1/3}$), with the horizontal F-factor over 1 km and a comparison of each landing's 200-ft band headwinds with the landings of the previous 45 minutes. The new alert levels are *WS advisory* (≥ 10 kt), *WSA* windshear alert (≥ 15 kt) and *MBA* microburst alert (≥ 30 kt loss), and every landing gets a stored windshear report. This document keeps the description of v1 so that observations and screenshots made with it — and statements referring to it — can still be understood.

1 Why the v1 algorithms were retired

A restudy on 24 October 2026 replayed the six algorithms on the stored raw replies of 3 084 landings at Helsinki-Vantaa (9–23 October 2026). The weather was benign: the highest METAR wind was 17 kt and no windshear was reported, so every alert was either a false alarm or the normal change of the wind with height. Share of approaches (Pairwise: of aircraft pairs) that would have reached **Warning (≥ 15 kt)**:

Algorithm	Warning in benign weather	Main reason
Energy	97.9 %	Uses ground speed + altitude/100; the planned descent and deceleration of every approach look like an energy loss (1 kt ≈ 12 ft at approach speed, not 100 ft).
Gradient	13.3 %	Largest difference of any two points 200–3000 ft apart (up to ~17 km of path): measures the normal increase of the wind with height; noise always raises the maximum.
Pairwise	7.6 % of pairs	Two different airframes up to 2000 ft (≈ 6 NM) apart: again the normal wind profile, plus each aircraft's own TAS / heading bias.
Rate	3.8 %	Look-back of up to ~2000 ft / 3 NM without scaling by distance.
Kinematic	2.3 %	IAS − GS: IAS is not TAS (≈ 4 kt difference at 2000 ft falling to ≈ 1 kt at 500 ft), so the descent itself fakes a headwind change; F as the maximum over sub-windows.
Baseline	0.2 %	Mixes all heights below 2000 ft into one expected value.

Common issues: no scaling of the change by the distance over which it happens (all operational methods — HKO GLYGA ramps, the 1-km F-factor, LLWAS station spacing — tie the threshold to a length or a time); altitude differences instead of the distance along the path; loss and gain treated alike; one poll's wind value per aircraft. The corridor and established-final gating, the go-around detection and the GNSS position-freeze guard were sound and are kept.

The v2 detector, replayed on the same landings, raised no windshear alert (WSA) and 37 advisories (1.2 %); the advisories are mostly real low-level wind-profile cases such as the night inversion of 11 October 2026 (+17 kt over 3 km between 1800 and 1200 ft in successive landings), and its band comparison caught the wind change of 17 October 2026 09:04 UTC on runway 22L (25–29 kt less headwind at 800–1200 ft than in the landings ten minutes earlier, seen by three aircraft in a row).

Source: *Low-level windshear from passively received Mode-S EHS — literature status, a review of the six algorithms in MODE-S Wind, and a noise-floor and false-alarm test on 3 084 Helsinki-Vantaa landings*, MODE-S Wind restudy, 24 October 2026 (with Shun & Chan 2008; Hon & Chan 2021; Lewis et al. 1994; ICAO Doc 9817).

2 The v1 implementation in brief

Item	v1 (static/js/windshear.js, 23 Oct 2026)
Where it ran	In the browser of each viewer, only while the Windshear page was open; one algorithm at a time, chosen per browser.
Input	/api/windshear/state every 3 s: each corridor aircraft's latest heading-calibrated wind (one BDS 5,0 + 6,0 pair), ground speed, IAS, altitude.
Headwind	wind speed × cos(wind direction − runway heading); runway headings 047° / 227° / 152° / 323° (true).
Levels	Monitor ≥ 10 kt · Warning ≥ 15 kt · Alarm ≥ 25 kt (magnitude, loss or gain).
Gates	≥ 6 corridor wind samples (establishment); 2 consecutive polls (confidence); grey (NONE) and stale-wind aircraft skipped; Pairwise / Kinematic only with glideslope status ON.
Pairwise	Pairs 200–2000 ft apart on the same runway.
Gradient	Wind history points ≥ 400 ft or 0.5 NM apart (≤ 40), max $\lvert\Delta HW\rvert$ over pairs 200–3000 ft apart.
Energy	E = GS + alt/100 over 45 s; loss ≥ 10 flagged.
Rate	Current headwind vs median of the oldest three of the last six stored points.
Baseline	Vector-mean wind below 2000 ft of the landings of the last 30 min (≥ 5 obs).
Kinematic	IAS − GS, medians of the first / last three samples of 45 s; F = max over 10–20 s sub-windows; optional F gate 0.05–0.15.
Log	In the browser only (50 entries, 60-s de-duplication), lost on page reload.

Appendix — README text of the v1 windshear detection (as of 23 October 2026)

Reproduced from README.md of 23 October 2026 (sections *Windshear Alert*, *Windshear detection* and *What is F-factor?*), unchanged apart from formatting.

Page note (as of 23 Oct 2026)

> **Keep the page open for best results.** All client-side history buffers — wind barb accumulation, kinematic IAS−GS differential history, the windshear event log, and the go-around log — exist only in the browser tab running the page. Navigating to another page or closing the tab clears these buffers entirely. When the Windshear page is reopened, it starts fresh: aircraft currently on approach will appear immediately, but any history built up during the previous session (wind profiles along the approach path, earlier windshear events, go-around events from before the page was opened) is gone. If you are monitoring an active approach sequence or want to study the windshear log over multiple arrival waves, keep the Windshear page as a dedicated open tab.

Windshear Alert + Today's Statistics

The bottom-right area is split into two panels side by side (55 % / 45 %).

Left — Windshear Alert log: maintains a compact timestamped log of all detected windshear events and go-around detections during the current session, newest first. Each entry is a single line showing time, algorithm badge, runway, magnitude, and callsign(s). Hovering over an entry shows a full tooltip with algorithm, altitude band, individual headwind components, and exact timestamp. Events are deduplicated per algorithm, runway, and aircraft within a 60-second window. The log is cleared by the **Clear** button or on page refresh and does not persist to the database.

Windshear detection

Windshear detection is controlled from the **Windshear Alert header bar**, which contains five inline controls: the **OFF/ON toggle button**, an **algorithm dropdown** (`Pair / Gradient / Energy / Rate / Baseline / Kinematic`), the **alert level** dropdown, the **Kinematic F-factor gate** dropdown, and the **Clear** button. Detection is **OFF by default** to allow monitoring of approach patterns before trusting automated alerts.

Six independent detection algorithms are available from the dropdown. Switching algorithm takes effect instantly and re-runs detection against the current aircraft set without waiting for the next poll. Only one algorithm is active at a time. Hovering over the dropdown shows a one-line description of each option.

Shear direction: every detected event is classified as `▼LOSS` (headwind decreasing — the operationally hazardous case) or `▲GAIN` (headwind increasing). The label appears in the alert banner, the ILS-profile canvas zone label, and the event log compact line. Severity thresholds (monitor / warning / alarm) are unchanged and based on the magnitude of the change regardless of direction.

Noise reduction: the Kinematic algorithm uses a 3-sample median on both the reference and the current measurement window, the Rate algorithm on the reference window (compared with the aircraft's current headwind) before computing the headwind differential. This suppresses single-sample transients (momentary IAS spikes, noisy BDS 6,0 decodes) without adding meaningful detection latency — three samples at the 3-second poll rate represent ~9 seconds of data.

Establishment gate: no algorithm can fire until an aircraft has accumulated at least 6 valid observations inside the corridor (≈15–20 seconds at the 3-second poll rate). This prevents false detections from the first wind snapshot, which is computed during the ILS intercept roll-out when the aircraft is transitioning from the turn onto final and the BDS 5,0/6,0 data is at its least stable.

Grey (NONE) aircraft are excluded: every algorithm skips aircraft whose current `meteo_source` is `NONE`, and their groundspeed / IAS samples are not added to the Energy and Kinematic buffers. An aircraft that goes grey during an intercept turn therefore cannot trigger an alert from a stale headwind value. This also means Energy and Kinematic only evaluate aircraft that currently have a valid computed wind.

All algorithms use the same headwind component formula:

```
headwind (kt) = wind_speed × cos(wind_direction − runway_heading)
```

Positive values represent a headwind; negative values represent a tailwind. The runway heading used is the approach heading from the tracker's runway table (true headings, e.g. 047° for RWY 04L/04R, 227° for RWY 22L/22R), matching the true wind direction from the wind calculation.

Events are classified into three severity levels based on headwind change magnitude:

Level	Threshold	Colour	Meaning
Monitor	≥ 10 kt	Blue	Informational — sub-threshold variation worth watching
Warning	≥ 15 kt	Amber	ICAO windshear threshold — operationally significant
Alarm	≥ 25 kt	Red	Severe windshear — significant aircraft performance impact

A **confidence gate** requires 2 consecutive poll cycles (≈ 6 seconds) both detecting the same event before it is promoted to the log or banner. This eliminates single-poll false positives from transient data artefacts without meaningfully delaying alerts for real sustained shear.

The **alert level selector** dropdown in the log header (`Mon ≥10kt` / `Warn ≥15kt` / `Alarm ≥25kt`) controls the minimum severity that activates the banner and flight strip WS badge. The log always shows all three levels regardless of this setting. The preference is saved across sessions. Default is Warning.

When a confirmed shear event meets the active alert level:

- An **alert banner** appears at the top of the page with the algorithm name, runway, altitude band, and aircraft information
- Affected flight strips get a pulsing **WS badge** (blue = monitor, amber = warning, red = alarm) — monitor badges do not pulse
- A coloured **horizontal band** (blue / amber / red) is drawn on the ILS profile canvas between the relevant altitudes
- The event is appended to the **windshear event log** with a coloured algorithm badge, timestamp, magnitude, gradient direction, and aircraft detail; Kinematic entries additionally show an **F-factor** value (e.g. `F=0.12`) — see What is F-factor? below for a full explanation

The **Pairwise** and **Kinematic** algorithms only include aircraft with GS status **ON** (within ±300 ft of the QNH-corrected glideslope), preventing false alerts from aircraft still intercepting the glideslope from above or below; the other algorithms rely on the establishment gate.

Turning the toggle OFF immediately clears all active alerts. Previously logged events remain visible until cleared manually. All data is held in RAM and does not persist across page refreshes.

Algorithm 1 — Pairwise (classic ICAO method)

Requires: ≥ 2 aircraft simultaneously on the same approach, each with ≥ 6 wind history observations (establishment gate).

The Pairwise algorithm is the classical windshear detection technique defined in ICAO Doc 9817 and used operationally by airport LLWAS (Low-Level Windshear Alert Systems). It compares the headwind component between two simultaneously tracked aircraft on the same ILS corridor, one higher and one lower on the glideslope.

For every pair of corridor aircraft within a 200–2 000 ft altitude separation window, the algorithm computes:

```
ΔHW = headwind(upper aircraft) − headwind(lower aircraft)
```

A large |ΔHW| reveals the presence of a wind shear layer between the two altitude levels. The 200 ft minimum separation prevents noise from aircraft at nearly the same level; the 2 000 ft maximum limits comparison to physically adjacent approach segments.

Limitation: requires two aircraft on approach simultaneously. During low-traffic periods this algorithm produces no output even when real shear is present. It is most powerful during busy arrival sequences with closely stacked aircraft.

Algorithm 2 — Gradient (single-aircraft wind history)

Requires: ≥ 6 stored wind observations for the same aircraft (establishment gate).

The Gradient algorithm examines the wind history accumulated for a single aircraft during its approach. As an aircraft descends from 3 000 ft to the threshold, the system stores up to 40 wind snapshots at intervals of ≥ 400 ft altitude change or ≥ 0.5 NM distance. The algorithm then finds the pair of observations — within a 200–3 000 ft altitude band — with the largest headwind difference.

This directly measures the **vertical wind gradient** (dHW/dz) in the low-level approach environment. A gradient exceeding ~15 kt over 1 000 ft of altitude is operationally significant and is indicative of a wind shear layer within the approach path.

Advantage over Pairwise: works with a single aircraft and accumulates evidence progressively as the aircraft descends through the wind field. Effective during single-runway VMC traffic when only one aircraft is on approach at a time.

Limitation: detection sensitivity increases with each new wind observation. The algorithm produces no output until the establishment gate is satisfied (< 6 valid wind points) and improves in accuracy as the aircraft continues its descent.

Algorithm 3 — Energy (total energy trend, GPWS-inspired)

Requires: ≥ 6 groundspeed + altitude observations within the last 45 seconds (establishment gate).

The Energy algorithm tracks a total mechanical energy proxy for each approach aircraft over a sliding 45-second window. The proxy is:

```
E = groundspeed (kt) + altitude (ft) / 100
```

On a stabilised 3° ILS approach at ~140 kt groundspeed, the aircraft descends roughly 318 ft per NM. Geometry and typical approach speeds mean that ~100 ft of altitude corresponds to approximately 1 kt of equivalent kinetic energy along the approach path. On a stable approach, E remains approximately constant as the aircraft trades altitude for forward speed at a predictable rate.

A rapid decrease in E signals that the aircraft is losing more total energy than the normal glideslope geometry predicts — the classic kinematic signature of a microburst or strong headwind loss event. A drop of ≥ 10 kt-equivalent in 45 seconds is flagged (Monitor); ≥ 15 is Warning and ≥ 25 Alarm, as for the other algorithms.

This algorithm is inspired conceptually by the energy-rate monitor in airborne EGPWS (Enhanced Ground Proximity Warning System) devices, which monitor the rate of total energy change to detect abnormal energy loss states during approach.

Advantage: the energy trend itself is groundspeed-based — no wind vector is needed for the calculation. (Aircraft without a valid computed wind are nevertheless skipped, see *Grey (NONE) aircraft are excluded* above.)

Limitation: any unrelated groundspeed fluctuation (e.g. temporary speed adjustment on ATC instruction) can trigger a false alarm. Works best with stable, consistent groundspeed data.

Algorithm 4 — Rate (headwind rate of change)

Requires: ≥ 6 stored wind observations and a current headwind value (establishment gate).

The Rate algorithm compares the aircraft's current headwind component to a reference taken from its recent wind history (median of the oldest up to three of the last 6 observations). Unlike the Gradient algorithm, it is not altitude-filtered — it detects any headwind change along the approach path, whether driven by altitude-related wind structure or by horizontal passage through a shear zone.

A large headwind change over a short segment — regardless of the altitude separation — indicates that the aircraft has rapidly entered a different wind environment. This catches purely horizontal or time-based wind shifts that develop at a fixed altitude level, including the leading edge of a microburst outflow where the shear layer may be nearly horizontal near the surface.

Advantage: sensitive to rapid temporal wind changes that have little altitude variation. Particularly useful on short final (< 500 ft) where altitude change between observations is small but wind speed can still shift dramatically.

Limitation: uses a short rolling window, so sustained slow changes may not accumulate to the threshold. More prone to transient false alarms than the altitude-separated algorithms.

Algorithm 5 — Baseline (historical approach deviation)

Requires: ≥ 5 low-altitude wind observations from completed approaches in the last 30 minutes.

The Baseline algorithm constructs a reference wind from recent completed approaches and compares each active corridor aircraft's current headwind to what was expected based on that reference.

When an approach aircraft disappears from the tracker (landed or left the corridor), the system harvests all of its stored wind observations below 2 000 ft and adds them to a rolling 30-minute buffer. The algorithm then vector-averages all observations in this buffer — using U/V component decomposition to handle the 360°/0° directional wrap correctly — to produce a background wind direction and speed. The expected headwind component for the current runway is derived from this averaged wind:

```
baseline_HW = avg_wind_speed × cos(avg_wind_dir − runway_heading)
```

If a current corridor aircraft's headwind deviates from `baseline_HW` by ≥ 10 kt (Monitor), ≥ 15 kt (Warning) or ≥ 25 kt (Alarm), shear is flagged — like the other algorithms, only for aircraft established in the corridor with at least the minimum number of corridor wind samples and valid meteo.

Physical basis: the baseline represents the background low-level wind field sampled by multiple recent aircraft on the same approach path. A large deviation for the current aircraft suggests that the wind environment has changed sharply since the baseline was established — either spatially (a localised shear zone has developed) or temporally (a frontal passage or microburst onset has changed the surface wind since the last landing).

Advantage: the most context-aware of the six algorithms — it adapts to the actual recent wind environment at the airport rather than using a fixed reference. It can detect wind changes that develop gradually between arrival waves.

Limitation: requires landing traffic in the preceding 30 minutes to populate the baseline buffer. The algorithm is silent at the start of a session or after a long traffic gap. Also assumes the baseline is representative of the current runway and direction, which may not hold perfectly when runway direction changes between the baseline and current approaches.

Algorithm 6 — Kinematic (IAS − GS differential rate)

Requires: a single aircraft in the ILS corridor with BDS 6,0 IAS data available.

The Kinematic algorithm detects windshear by monitoring the rate of change of the difference between Indicated Airspeed (IAS) and GPS groundspeed (GS) over a 45-second sliding window.

At low altitude, air density is close to sea-level ISA conditions, so IAS ≈ TAS (True Airspeed). The identity `IAS − GS ≈ headwind component along track` therefore holds without requiring any wind direction decoding. If the aircraft is flying into a headwind the differential is positive; a tailwind makes it negative. A sudden change in this differential is a direct kinematic measurement of a windshear encounter:

```
differential(t) = IAS(t) − GS(t)
diffOld = median(differential, oldest 3 samples in 45 s window)
diffNew = median(differential, newest 3 samples in 45 s window)
delta   = |diffNew − diffOld|
```

The median of 3 samples (rather than a single raw point) is used at each end to suppress single-poll IAS transients or noisy BDS 6,0 decodes before differencing. At the 3-second poll rate, three samples represent approximately 9 seconds of data.

If `delta ≥ 10 kt` the event is recorded at **Monitor** level (sub-threshold, informational). `≥ 15 kt` is **Warning**; `≥ 25 kt` is **Alarm**.

Physical basis: the approach phase is the only flight phase where a sudden change in `IAS − GS` can reliably be attributed to a wind shear and not to a deliberate speed change, because: - the aircraft is flying a stabilised approach at a fixed target IAS; - any large unintended IAS change is caused by an aerodynamic disturbance (gust, shear, microburst), not a throttle input; - GS reflects the aircraft's inertial motion, which lags momentarily behind the aerodynamic change, creating a measurable differential.

A microburst headwind-loss encounter produces a rapid decrease in the differential (IAS drops faster than GS responds), while a headwind-gain event produces the opposite.

Advantage: the most dataflow-simple of all algorithms — uses only two raw Mode S fields (BDS 6,0 airspeed, ADS-B groundspeed) with no wind vector reconstruction. Works with a single aircraft. No runway heading or wind direction needed.

Limitation: requires BDS 6,0 Indicated Airspeed to be broadcast by the aircraft, which most modern jets do but is not mandatory. The IAS ≈ TAS approximation breaks down above ~6 000 ft (density altitude effect), but the algorithm is gated to GS-status-ON aircraft which are already on the glideslope well below that altitude.

Why Kinematic is the recommended algorithm for MODE-S EHS monitoring:

The other five algorithms all ultimately depend on computing a wind vector from BDS 5,0 (track, groundspeed, TAS) and BDS 6,0 (magnetic heading), then projecting it onto the runway axis as a headwind component. That chain has several vulnerable links specific to Mode-S data:

- **Magnetic heading errors** — BDS 6,0 magnetic heading is encoded at ~0.18° resolution (90/512°) and some transponders carry small per-aircraft biases from the compass installation. Small heading errors produce proportionally large wind direction errors at approach speeds.
- **Magnetic declination** — converting magnetic heading to true heading requires an accurate declination. The position-based WMM value removes the main error, but any residual difference between the model and the aircraft's own magnetic variation table still shifts computed wind vectors systematically, affecting pairwise, gradient, rate, and baseline equally.
- **BDS 6,0 availability and quality** — IAS and magnetic heading are in the same BDS 6,0 register; a transponder broadcasting a bad magnetic heading also corrupts the wind vector even when IAS itself is clean.
- **Multi-aircraft requirements** — Pairwise requires two aircraft simultaneously on the same corridor at different altitudes, which is not always available. The other single-aircraft algorithms need a long wind barb history to accumulate enough altitude range or time span to build a meaningful comparison.

Kinematic sidesteps all of this. The `IAS − GS` differential is computed from two independent data sources: BDS 6,0 IAS (airspeed sensor) and ADS-B groundspeed (GPS/GNSS). Neither requires a heading, a declination constant, or any wind vector calculation. A change in the differential is a direct aerodynamic measurement — the aircraft itself is the sensor. No amount of magnetic heading noise, declination drift, or

runway geometry uncertainty can corrupt it. At EFHK, where the GPS jamming environment from the east adds an additional source of position and velocity noise for other algorithms, this independence is particularly valuable.

The practical consequence is that Kinematic produces the fewest false events from data-quality artifacts, while remaining sensitive to real shear. The 3-sample median filter (added to smooth single-poll IAS transients) and the corrected JAWS-reference F-factor gate give further control over the signal-to-noise trade-off.

What is F-factor?

Kinematic log entries display an F-factor value alongside the kt delta (for example `18 kt · F=0.12`). F-factor is a dimensionless number that answers the question: *how fast is the headwind changing relative to gravity?*

```
F = (headwind change in m/s) / (time window in seconds) / 9.81
```

Gravity (9.81 m/s²) is used as the normaliser because aircraft performance is ultimately governed by it — a headwind loss creates a sink rate, and that sink rate competes with the aircraft's ability to maintain the glideslope. An F-factor of 0.1 means the headwind is decaying at one-tenth of gravitational acceleration, which is the point where most approach aircraft start to struggle to maintain energy.

Reference thresholds from NASA/FAA JAWS research:

F-factor	Interpretation
< 0.05	Negligible — normal approach variation
0.05–0.10	Noticeable — pilot may sense it
≥ 0.10	Operationally significant — performance impact likely
≥ 0.15	Severe — significant control input needed

These thresholds come from the Joint Airport Weather Studies (JAWS) programme which established F-factor as the standard hazard metric used in airborne windshear warning systems (GPWS/EGPWS).

Why F-factor adds information beyond the kt value: two events could both show `18 kt Warning` but have very different F-factors depending on how quickly the 18 kt was lost. An aircraft that lost 18 kt of headwind over 40 seconds (`F ≈ 0.024`) is a very different situation from one that lost it over 8 seconds (`F ≈ 0.12`). The kt value tells you the magnitude; the F-factor tells you the rate, which is what determines whether the crew had time to respond.

F-factor is displayed as supplementary information in the log. Severity classification is still driven by the kt thresholds, but the **Kinematic F-factor gate** dropdown (`F: Off / F ≥0.05 / F ≥0.08 / F ≥0.10 / F ≥0.15`) in the log header lets you suppress Kinematic events whose F-factor is below a chosen minimum. This is useful for research — setting the gate to `F ≥0.05` silently discards the slow background drift events (typically F=0.02–0.03) that represent normal approach variation according to the JAWS thresholds, while preserving genuine rapid shear events. The gate is automatically greyed out when any other algorithm is selected, since F-factor is only computed for Kinematic. The preference is saved across sessions. Default is Off (no filtering).

F-factor reference window: F is computed by scanning all 10–20 second sub-windows within the 45-second detection history and reporting the maximum. This matches the JAWS 1 km reference distance (≈15 s at typical approach speed). The 45-second detection window that determines whether an event fires is unchanged — only the F-factor value reported is affected. The F-gate thresholds in the table above are therefore correctly calibrated against the published JAWS values.